Jegannarayanan Sivakasiulaganathan

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Education

University of Massachusetts Lowell Expected May 2027
M.S. Computer Science (GPA: 3.7/4.0)
Relevant Coursework: Compiler Engineering, Effective Functional Programming (`Haskell`), Graduate Algorithms, Operating Systems I, Deep Learning (LLMs/LVMs).

University of Massachusetts Amherst Graduated May 2018
B.S. Computational and Applied Mathematics

Research and Compiler Projects

Master's Thesis: Lean to MLIR Tensor Lowering *(in progress)* September 2026–Present

- Investigating a lowering path from high level tensor operations in a Lean tensor library to MLIR's `linalg` dialect, using matrix multiplication as an initial case study.
- Analyzing Lean compiler representations (`Expr`, `IR`, `LCNF`) to decide on which representation to lower to MLIR.
- Wrote corresponding Lean and MLIR matrix multiplication examples and examined semantic preservation requirements.

Haskell Project: Python AST Traversal and Translation

- Project that used a Python parser to generate an AST, then implemented Haskell AST transformations to convert it into a custom AST.
- Designed the custom AST to represent a small subset of Python and support eventual translation into MLIR using a library called `mlir-hs`.

Toy Programming Language Interpreter in C++

- Implementing an interpreter in C++ for a toy language to explore lexing, parsing, and interpreter architecture.
- Completed the lexer and began implementing the parser without `flex` or `Bison`, working through architectural decisions involving when and where to report errors as well as statement and expression parsing details.
- Explored C++ memory ownership and object lifetimes involving heap allocation and `std::unique_ptr`.

Professional Software Development Experience (7 Years)

KBRA Dresher, PA (Remote)
Software Developer (promoted from Associate Software Developer) *Aug 2022–July 2025*

- Independently led a tech debt elimination initiative, coordinating engineers, maintaining technical documentation, and advocating for the migration from an unsupported PHP framework to C#/.NET services.
- Developed C# ASP.NET Core backend services, worked with Hot Chocolate GraphQL schemas, and contributed to React frontend features while collaborating with staff engineers.
- Independently investigated C#/.NET technologies beyond immediate feature requirements, including but not limited to: Hot Chocolate GraphQL, the ASP.NET Core HTTP request pipeline, and C# record constructors. Applied this deeper understanding to make informed implementation decisions and better support the team.

Aspen Technology Bedford, MA
Software Developer (previously Software Developer I) *May 2018–Aug 2022*

- Independently diagnosed and resolved defects working on multiple production C# desktop applications, reproducing customer reported issues, tracing root causes, and implementing fixes.
- Progressed from Software Developer I to Software Developer in July 2021.

Technical Skills

Programming: C++, Python, Haskell, Lean 4, Fish, C#
Compiler Engineering: MLIR, Parsing, AST Design and Transformation
Software and systems: Linux, Git, Docker